

DEV_IMPL_GUIDE - FOUNDATION_CACHE

1. Purpose

This guide explains how to implement the Redis/cache foundation.

Use it with FOUNDATION_CACHE.md.

This is a provisioning/runtime-setup guide. It is intentionally written as infrastructure setup steps rather than a business request execution flow.

2. What You Are Building

You are deploying Redis for non-authoritative cache use:

- Reference data cache.
- Session/runtime cache where approved.
- Rate-limit or short-lived computed data.

Redis must not be the source of truth for subscription, billing, fulfillment, or workflow state.

3. Step 1 - Provision Redis

Recommended medium setup:

```
4 Redis nodes
Redis Cluster enabled
replication configured by shard
```

Use the sizing and HA behavior from the DD.

4. Step 2 - Define Cache Names

Use explicit key prefixes:

```
sophix:package:<operator>:<packageRef>
sophix:homepass:<operator>:<homepassId>
sophix:authz:user:<userId>
sophix:rate:<clientId>:<window>
```

Rules:

- Every key must have a TTL unless explicitly approved.
- Include operator scope in keys.
- Do not cache secrets in plain text.

5. Step 3 - Implement Cache Wrapper

Applications should use a wrapper:

```
CacheClient.getOrLoad(key, ttl, loader)
CacheClient.invalidate(key)
CacheClient.setJson(key, value, ttl)
```

Do not scatter raw Redis calls across business code.

6. Step 4 - TTL Policy

Suggested TTLs:

Data	TTL
Package snapshot	5-15 minutes
HomePass sellability snapshot	1-5 minutes
Authz/group lookup	1-5 minutes
Rate-limit counters	Window duration

Data	TTL
Temporary workflow lookup	Less than operation timeout

7. Step 5 - Invalidation

Preferred invalidation:

- Listen to catalog/config events.
- Invalidate affected keys.
- Reload on next request.

Fallback:

- TTL-only expiration.

8. Step 6 - Monitoring

Alert on:

Metric	Alert
Redis node down	Immediate.
Memory usage	Above 80 percent.
Evictions	Above zero sustained.
High latency	Above SLA threshold.
Replication issue	Any sustained issue.

9. AWS Option

If using AWS:

- Use ElastiCache for Redis.
- Enable Multi-AZ.
- Use private subnets.
- Enable encryption if supported by selected mode.
- Use security groups to restrict access.

10. Tests

Test:

- Cache miss loads from source.
- Cache hit avoids source call.
- TTL expires.
- Invalidation works.
- Source-of-truth fallback works when cache is unavailable.

11. Common Mistakes

- Storing authoritative state in Redis.
- No TTL on reference cache.
- Missing operator code in keys.
- Letting Redis outage break critical write workflows when fallback exists.